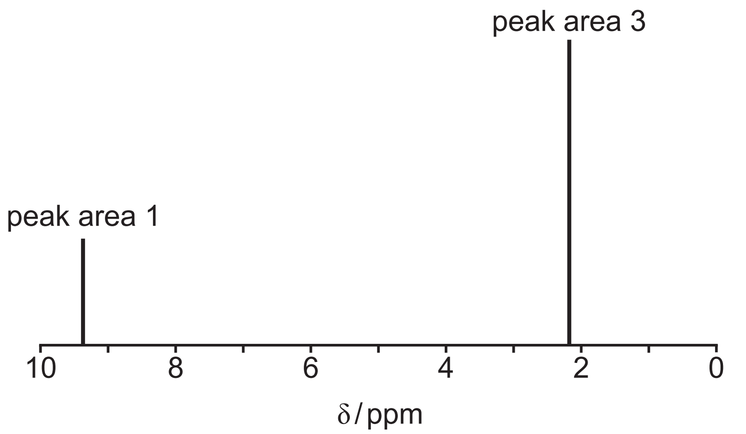
10. (a) Compound **G** contains only carbon, hydrogen and oxygen.

It has a molar mass of 72 g mol^{-1} of which 50.0% is carbon. The compound reacts positively with Tollens' reagent and gives a yellow solid when treated with alkaline iodine solution. It reacts with sodium tetrahydridoborate(III) to give a new compound which has a molar mass of 76 g mol^{-1}.

The high resolution ^{1}H NMR spectrum of compound **G** is shown below.



Examiner only

Use **all** of this information to deduce a structure for compound **G**.

You should comment on how each piece of data has helped you to deduce the structure. [6 QER]

Examiner only

(b) The absorption spectrum of compound **G** shows a maximum absorption at 450 nm.

Light energy measured in kJ mol^{-1} (E) is related to wavelength (λ) by the equation

$$E = \frac{\text{constant}}{\lambda}$$

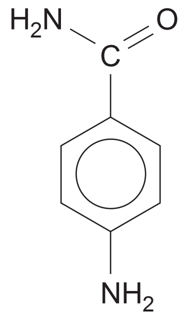
The energy of light of wavelength 656 nm is 183 kJ mol^{-1}.

Calculate the energy of the maximum absorption of compound **G** at 450 nm. [2]

Energy = ... kJ mol^{-1}

Examiner only

(c) The formula of 4-aminobenzamide is



(i) When this compound is warmed with excess aqueous sodium hydroxide, only the amide group reacts.

I. Give the structure of the organic product. [1]

II. State the nature of the attacking reagent and explain why the NH_2 group bonded to the benzene ring is not removed. [2]

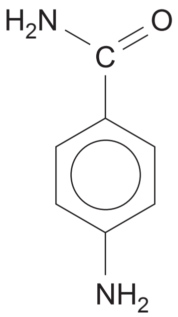
....

....

....

(ii) Both the functional groups in 4-aminobenzamide are attacked by nitric(III) acid at room temperature, giving 4-hydroxybenzoic acid, nitrogen gas and water as the products.

Complete the equation for this reaction. [2]



+ .... HNO_2 $\longrightarrow$ + ... N_2 + ... H_2O

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